REFINEMENT vs. REFRESH

Where should a robot policy spend its inference compute?

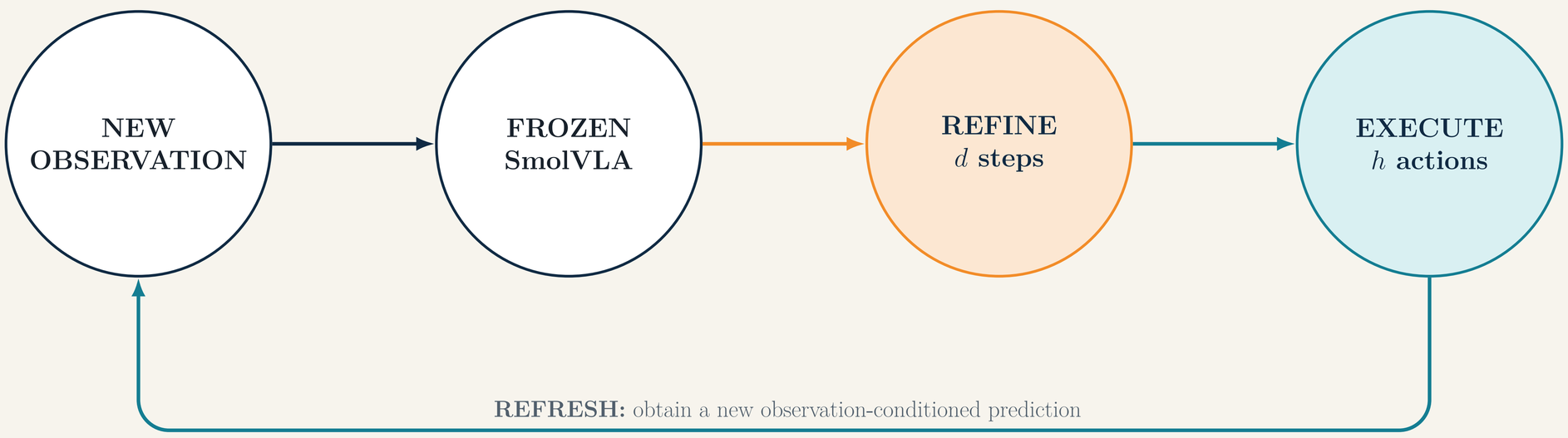
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A chunked policy has two ways to react:

Improve the current prediction, or act briefly and plan again from a fresher observation.

THE DEPLOYMENT CHOICE: how much d, how much h?



MAIN HELD-OUT COMPARISON

Less refinement improved closed-loop success

Same two-action execution prefix; eight Goal and Object tasks excluded from configuration selection.

2 denoising steps

139 / 160 successes

5 denoising steps

122 / 160 successes

+10.6 pp

descriptive 95% interval
+4.4 to +16.9 pp

−5.9%
arithmetic per control step with 2 instead of 5 denoising steps

82.1%
of generation arithmetic is independent of denoising count

−60.7%
arithmetic per attempt for $h = 5$ versus $h = 2$, at $d = 2$

HOW WE TESTED IT

12 configurations
4 Spatial tasks
480 discovery attempts

7 frozen configurations
8 held-out tasks
1,120 confirmation attempts

One frozen checkpoint
Synchronous LIBERO
Paired evaluation

TAKEAWAY

More denoising does not necessarily improve closed-loop control.

Evaluate refinement and replanning together, using both task success and compute.

INTERPRET WITH CARE

Similar observed success for $h = 2$ and $h = 5$ does not establish equivalent quality. Results cover one checkpoint and synchronous simulation.

github.com/amtellezfernandez/refinement-refresh-robot-policies